

Jonathan Lee

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EDUCATION

Northwestern University

B.S. in Computer Science and Mathematics (Pure Mathematics concentration)

Evanston, IL

Expected June 2027

- GPA: 3.84
- CS Honors Track and Research Track
- McCormick Dean's List (5 of 6 quarters)

University of Kansas

Undergraduate coursework, transferred to Northwestern

Lawrence, KS

August 2023 – May 2024

- GPA: 4.00
- Albert F. and Barbara Kile Zernickow Mathematics Scholarship

RESEARCH EXPERIENCE

Researcher, Cryptography

Advisor: Prof. Xiao Wang, Northwestern University

January 2026 – Present

Evanston, IL

- Honors thesis (in progress): naysayer proofs for post-quantum signature verification on blockchains
- Selected parameters for and benchmarked a lattice-based commitment construction (holomorphic Merkle trees) on Northwestern's Moore compute cluster
- Studied blockchain address poisoning attacks and countermeasures

Researcher, Formal Methods & Algebraic Structures

Advisor: Prof. Dmitrii Pasechnik, Northwestern University

September 2025 – August 2026

Evanston, IL

- Formalized strongly regular graphs in the Rocq proof assistant, proving core parameter identities and complement closure using the MathComp library
- Implemented certified constructions of Paley and Hamming (rook's) graphs with verified extraction to OCaml, and built a reproducible test pipeline for validating instances

TEACHING

Peer Mentor (TA)

Northwestern University

Fall 2026

Evanston, IL

- CS 212: Mathematical Foundations of Computer Science (Discrete Math)
- CS 180: new introductory programming course, assisting with course development

TALKS & PRESENTATIONS

Character Theory of Finite Groups

Spring 2026

- Directed Reading Program capstone talk concluding a two-quarter sequence. Covered material from Isaacs' *Character Theory of Finite Groups*, including Clifford theory, the Frobenius–Schur indicator, and induction formulas

Representation Theory of Finite Groups

Winter 2026

- Directed Reading Program talk on representation-theory foundations (via *A First Course in Representation Theory*), which led into the character-theory talk above

Formalization of Strongly Regular Graphs in Rocq

Fall 2025

- Presented at Northwestern undergraduate research expo, demonstrating certified constructions and verified extraction pipeline

SELECTED COURSEWORK

Cryptography & Security: Introduction to Cryptography, Advanced Topics in Modern Cryptography (graduate), Blockchains & Decentralization

Computer Science: Graduate Algorithms (in progress), Design & Analysis of Algorithms, Theory of Computation, Quantum Computing (in progress), Programming Languages, Programming Massively Parallel Processors, Machine Learning

Mathematics: Abstract Algebra (3 quarters), Complex Analysis, Real Analysis (in progress), Linear Algebra: Second Course

Directed Reading: Representation Theory, Character Theory, Elliptic Curves (Summer 2026)

Logic: Elementary Logic II

INDUSTRY EXPERIENCE

Software Engineering Intern June 2026 – August 2026
AlphaSense Chicago, IL

- Built features on the Client & Expert Experience team across two large production repositories in React, Vue, and Ruby behind a GraphQL middle layer
- Implemented end-to-end changes spanning GraphQL schema and resolvers through to the client

Full-Stack Software Engineering Intern June 2025 – April 2026
W.R. Berkley – Intrepid Direct Insurance Overland Park, KS

- Owned a Salesforce Batch Apex job automating the rollover of renewal coverage terms, eliminating per-policy manual entry
- Built round-robin account assignment logic and internal tooling in .NET Core, Angular, and Apex

Software Engineering Intern May 2024 – August 2024
WellSky Overland Park, KS

- Shipped UI and backend improvements to a clinical workflow platform serving 2000+ healthcare providers
- Implemented a patient management feature in Ext JS for pharmacy clients, enabling them to track patient medication adherence

PROJECTS

Galois Playground | *Python, React, SageMath, FastAPI*

- Web application using SageMath (via a FastAPI backend) to visualize group structure, subgroup lattices, and quotient groups

LEADERSHIP

President, Blockchain Group at Northwestern April 2026 – Present
Northwestern University Evanston, IL

- Revived the dormant group by re-registering it with the university, writing its constitution, and securing student activities funding (SAFC)
- Recruited three officers and run weekly member meetings
- Organize an exploratory member project on lattice-based signatures for blockchains

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, C#, Solidity, Ruby, Java, TypeScript, SQL, Apex

Mathematical Software: SageMath, GAP, Rocq/Coq, LaTeX

Tools & Frameworks: Git, Linux, CUDA, React, Vue, Angular, GraphQL, .NET, Node.js, Rails

PERSONAL INTERESTS

Rock climbing, recorder, basketball, game development